

BRAEDEN KEENA

Data Engineer • Recurring Pipelines • Data Integrity & Failure Detection

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I build data collection that keeps working, and I specialize in the failure that looks like success — a pipeline reporting a clean run while returning nothing. Every figure I publish is emitted by a committed script that refuses to write a null, because I once shipped a hand-typed number that was wrong by 5.6× for an unknown period. Live deployed web applications built end to end: API reverse-engineering, data pipelines, front-end, SEO, and daily automation. Backed by front-line operations leadership — including founding and running my own business — managing teams of up to 100 people. I learn whatever a problem requires, then build the thing.

SHIPPED PRODUCTS (SOLO-BUILT, END TO END)

WiFi CSI RF-Fingerprinting Sensor Array — ESP32 + Python DSP

2026 • Firmware / DSP

- Built a WiFi device-fingerprinting system end to end — ESP32 firmware in C / ESP-IDF, a Python DSP chain that pulls sampling-clock error out of raw subcarrier phase (RANSAC phase-slope fits, 1D Kalman drift tracking, Mahalanobis discrimination against per-source Gaussian models), and a classifier reaching 99.7% blind-holdout accuracy separating three transmitters on a chronological split. 43.4 GB of captures across 97 files and 60 recording sessions.
- Then spent two months trying to break it, and succeeded. Twenty-nine hypotheses carried to a verdict, thirteen of them my own working results, each retired by a control built to permit exactly that outcome — reference-beacon correction, longer averaging windows, tighter gates, blind clustering, feature fusion. Sixteen more written down as unresolved rather than quietly re-run.
- The system was separating rooms, not radios. Two different boards placed four feet apart became statistically indistinguishable at 0.2σ; one board carried between rooms moved 19.6σ. Propagation path had never been varied, so “which device is this” and “which room is this” had been the same question since July — a design flaw, not a bug, and nothing in 43 GB of captures recorded where any hardware sat.
- Feature fusion appeared to add +26 percentage points, winning 12 of 12 comparisons with a confidence interval excluding zero; a random number of identical variance reproduced the gain exactly at +0.00 pp. Thresholds and controls written to disk and dated before any analysis ran, chronological train/test splits never random, and six nights into a seven-night pre-registered replication whose primary prediction has failed every night it was tested.

findstorage.pages.dev — Self-Storage Price Tracker 2025 – 2026 • Founder / Developer • Sunset by choice

- Daily national advertised-price tracker: 4,664 stores across 43 states, 48 consecutive daily snapshots with zero gaps, 396,402 price changes and 148,870 promotion changes logged. 7-pass discovery scraper with safety rails that abort a publish if the store count drops below a floor or swings >10% run-over-run — an operator merger added ~1,100 stores overnight and the pipeline absorbed it with no manual intervention.
- Shut the project down on 25 August 2026 after re-reading the source’s terms of use, and published why rather than stopping quietly — killing a six-week analysis that a lawyer had already cleared, because the data underneath it was collected under terms I had not re-checked.
- Found and fixed four separate errors in my own published analysis before shutting down, including a coverage figure wrong by 5.6× that no script had ever recomputed. Every published number now traces to a named command.
- Nationwide ZIP3 market analysis with matched-pairs attribute pricing (isolated a ~18–22% climate-control premium via same-store/same-size comparison), a “renter leverage” score, and per-store price-history drill-down.

Public-Records Extraction — GAScout • easyIcE • KcPermits

2025 – 2026 • Solo-built

- Turn government systems never designed to be read programmatically into structured, deduplicated, ranked datasets. GAScout (gascout.pages.dev): \$152.34M in delinquent taxes and 451,798 records across four live Georgia counties in four source formats — a 9,915-page mainframe PDF, a weekly XLSX roll, a monthly PDF, and a scanned PDF read by OCR — normalized to one schema.
- 100% line accounting on the mainframe parse: 409,142 rows, zero dropped, 32 column-overflow swaps auto-corrected by a deterministic 5-column checksum, byte-identical output hash across machines, 90 automated tests. The remaining 155 counties render “ADAPTER PLANNED” with no synthetic figures anywhere on the site.
- easyIcE converts county health-inspection records into a recency-scored call list across five jurisdiction adapters and three source shapes (Socrata JSON API, server-rendered HTML with sequential IDs, Tyler portals); 279 tests, none

touching the network. Every run publishes a coverage report that declares its own degradation first — a partial run cannot be mistaken for a complete one.

nacup.us — Live World Cup 2026 Fan Hub

2026 · Founder / Developer

- Designed, built, and deployed a full-stack production site in 48 hours; reached first-page organic search traffic within 24 hours of launch with 700+ visitors on day one during a live global event.
- Multi-source data pipeline (Google Places API, RSS, news scraping, live scores REST API) with dedupe/merge logic, trust-tier scoring, and freshness decay ranking across 38 host-city feeds.
- Programmatically generated 80+ static SEO pages with JSON-LD structured data, sitemap, and Search Console integration; live-score polling with caching and fallback states.

TECHNICAL SKILLS

- **Languages:** Python (scraping, data pipelines, automation), SQL, JavaScript, C, HTML/CSS
- **Data:** ETL pipelines, entity resolution, deduplication, pandas, SQLite, Parquet, REST APIs, API reverse-engineering, Playwright, PDF and OCR extraction (Tesseract, pdfplumber), JSON data architecture
- **Testing & Reliability:** pytest, fixture-based contract tests, negative controls, coverage reporting, pre-registered thresholds, failure detection and alerting
- **DevOps:** Git/GitHub, CI/CD (GitHub Actions, Cloudflare Pages, Netlify), scheduled jobs, robots.txt and rate-limit compliance
- **Hardware/IoT:** ESP32 firmware (C / ESP-IDF), ESP-NOW & BLE, I2C peripherals, sensor networks, 3D printing & CAD (OpenSCAD)

OPERATIONS & LEADERSHIP EXPERIENCE

Front-line operations and team leadership — including founding and operating my own business — managing teams of up to 100 people across high-volume service environments, staffing, and vendor coordination.

Independent Technical Consultant

7/2026 – Present

- Build and ship production scrapers, data pipelines, and automation — turning messy public and web data sources into clean, scheduled, queryable datasets and dashboards (see shipped projects above).
- Contract work delivered across records consolidation and data visualization; 100% Job Success Score and Rising Talent status on Upwork as of September 2026.

Crossing Guard Operations Supervisor — All City Management Services (ACMS)

2023 – 2025

- Hired, trained, and managed 50 crossing guards across 33 sites — 66 posts staffed twice every school day — holding ~25% annual turnover in a category known for severe churn.
- Coordinated daily across five constituencies (county sheriff, school principals, parents, the district, and the guards) and solved same-day coverage through a deep bench of on-call relationships — every student returned home safely, zero students injured across my full tenure.

Founder / Sole Operating Partner — Honey Lake Motocross Park, Milford, CA

2016 – 2018

- Acquired and revived a ~500-acre desert motocross park dormant since 2007 — bought with two capital partners as the sole operating owner — and re-entitled it for racing after the prior owner had lost county approval and stripped the site: new survey, environmental review, a water well drilled under budget, and a heavy-equipment track rebuild.
- Directed a 120-person race-day crew built largely on in-kind pay with no payroll budget (peak ~6,500 attendees); designed and staffed the on-site medical program that executed a mid-event helicopter evacuation without incident.
- Held full P&L — pricing, sponsorship, staffing, and marketing — and founded the Corey Herring Memorial race, which continued annually through 2023; exited by selling my stake.

EDUCATION

GED — State of Kansas, 2014. Everything since: self-taught, project-driven, and shipped to production.